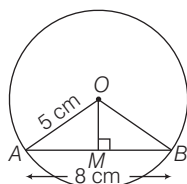


HINTS AND SOLUTIONS

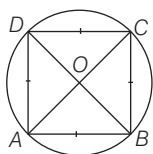
1. (c) Given, $OA = 5$ cm, $AM = \frac{1}{2} \times AB = \frac{8}{2}$
 $\Rightarrow AM = 4$ cm



In right angled $\triangle OMA$,
 $OM^2 = OA^2 - AM^2$
 [by pythagoras theorem]
 $= 5^2 - 4^2 = 25 - 16 = 25 - 16 = 9$
 $\Rightarrow OM = 3$ cm

2. (a) $\therefore \angle BOC = 2\angle A = 2 \times 60^\circ$
 (Angle subtended by an arc at the center is double the angle subtended by it at any point on the circumference of circle)
 $\Rightarrow \angle BOC = 120^\circ$

3. (d) A square has four equal side.
 $\therefore$ Each side subtends the same angle at the centre O.
 Let angle subtended be x° .

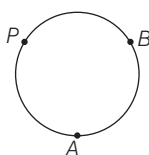


$$\text{So, } 4x^\circ = 360^\circ$$

$$x^\circ = \frac{360^\circ}{4} = 90^\circ$$

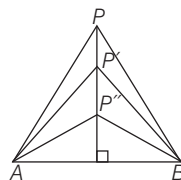
4. (b) Clearly, the point of concurrency of the perpendicular bisectors of the sides of a triangle is known as circumcentre.
5. (c) As, O is mid-point of diameter PQ and $\angle PRQ = 90^\circ$ (angle in semi-circle)
 So, $OS \parallel QR$ and $OS \perp PR$
 $\therefore \frac{PO}{PQ} = \frac{OS}{QR}$
 $\Rightarrow \frac{OS}{QR} = \frac{1}{2} \Rightarrow OS = \frac{1}{2} QR$
6. (d) $PQ = RS$ as chords equidistant from the centre are equal.
7. (a) The locus of the mid-points of all radii of a circle is a circle.
8. (c) The perpendicular bisectors of the sides of a triangle passes through the same point. The perpendicular bisectors are concurrent and point is called the circumcentre.

9. (a) Let P be the moving point.
 Then, $PA^2 + PB^2 = \text{constant}$



So, the locus of P is a circle.

10. (c) The locus of points equidistant from fixed points is a straight line which is the perpendicular bisector of the line joining the two fixed points.
 Here, A, B are fixed point and P is locus.

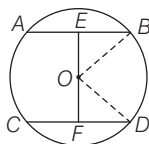


11. (c) Given that, $\Rightarrow \angle BOD = 50^\circ$
 $\Rightarrow \angle AOD = 180^\circ - \angle BOD = 130^\circ$
 [linear pair]

12. (a) Here, $\angle CAB = \angle CBD$
 [angles in alternate segments]
 and $\angle DAB = \angle CDB$
 [angles in alternate segments]
 $\Rightarrow \angle CAD = \angle CAB + \angle DAB$
 $= \angle CBD + \angle CDB$
 On adding $\angle CBD$ both sides, we get
 $\angle CAD + \angle CBD$
 $= \angle CBD + \angle CDB + \angle CBD$
 $= 180^\circ$ [angles of a triangle]

13. (d) Here, $BE = \frac{1}{2} AB = \frac{16}{2} \text{ cm} = 8 \text{ cm}$

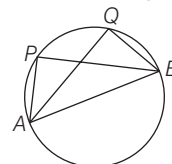
$OB = OD = 17$ cm (radii)



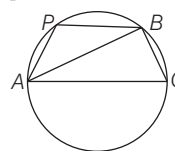
In right angled $\triangle OEB$,
 $OE = \sqrt{OB^2 - BE^2}$
 $= \sqrt{17^2 - 8^2} = \sqrt{289 - 64}$
 $= \sqrt{225} = 15$ cm
 $\therefore OF = EF - OE = (23 - 15) = 8$ cm
 In right angled $\triangle OFD$,
 $FD = \sqrt{OD^2 - OF^2}$
 $= \sqrt{17^2 - 8^2} = \sqrt{289 - 64}$
 $= \sqrt{225} = 15$ cm
 $\therefore CD = 2FD = 30$ cm

14. (c) There are two possibilities

Case I When P and Q are on the same side of AB. In this case $\angle APB = \angle AQB$ (angle in the same segment).



Case II When P and Q are on the opposite of AB. In this case PAQB is a cyclic quadrilateral.



So, $\angle APB + \angle AQB = 180^\circ$

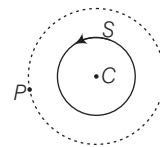
15. (b) Clearly, the locus of the centre of circles which passes through two given point is perpendicular bisector of the line joining the given points.

16. (c) As POQ is the diameter of the circle, therefore, $\angle PRQ$ and $\angle PSQ$ are two semicircles and $\angle PRQ = \angle PSQ$.
 $\angle PRQ = \angle PSQ = 90^\circ$

[angles in semicircle]

17. (a) $\angle COB = 360^\circ - (110^\circ + 90^\circ) = 160^\circ$
 $\therefore x = \angle CAB = \frac{1}{2} \angle COB$
 $= \frac{1}{2} \times 160^\circ = 80^\circ$

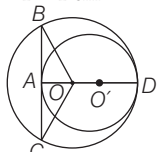
18. (b) If S is a circle with centre C and P be a movable point outside S, then the locus of P such that the tangent from P to S are of constant length is the circle through P with centre at C.



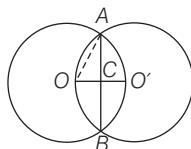
19. (a) $\angle CBF = \angle CDA$
 (angles in alternate segments)
 $\Rightarrow \angle CDA = 130^\circ$
 $\angle CDA + x = 180^\circ$ [linear pair]
 $x = 180^\circ - 130^\circ = 50^\circ$
 Hence, the value of x is 50° .

20. (c) $\angle ACB = \angle ADB = 32^\circ$
 [angle in same segment]
 $\angle ACD = \angle ABD = 50^\circ$
 $\therefore x = \angle BCD = \angle ACB + \angle ACD$
 $= 32^\circ + 50^\circ = 82^\circ$

21. (d) Here, $OB = OD = 3$ cm and
 $O'D = O'A = 2$ cm
 $OO' = OD - O'D = 3 - 2 = 1$ cm
 $\therefore OA = O'A - OO' = 2 - 1 = 1$ cm
 In $\triangle OAB$,
 $AB = \sqrt{OB^2 - OA^2}$
 [by pythagoras theorem]
 $= \sqrt{3^2 - 1^2} = \sqrt{9 - 1}$
 $= \sqrt{8}$ cm
 $\therefore$ Required length $= BC = 2AB$
 $= 2\sqrt{8}$ cm $= 4\sqrt{2}$ cm



22. (b) Here, let O, O' be the centres of the circles.
 As, the centre of each lies on the circumference of the other, therefore the two circles will have the same radius. Let the radius be r .
 $\therefore OC = O'C = \frac{r}{2}$



In right angled $\triangle OCA$, by pythagoras theorem

$$\begin{aligned} \therefore AC &= \sqrt{OA^2 - OC^2} \\ &= \sqrt{r^2 - \left(\frac{r}{2}\right)^2} = \frac{\sqrt{3}}{2} r \\ \Rightarrow AB &= 2AC = \sqrt{3} r \\ \therefore \frac{\text{Common chord}}{\text{Radius}} &= \frac{AB}{OA} = \frac{\sqrt{3}r}{r} \end{aligned}$$

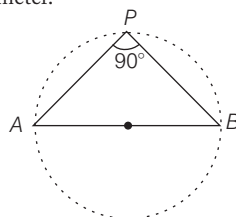
Hence, ratio of common chord to the radius is $\sqrt{3} : 1$.

23. (c) $\therefore \angle B + \angle D = 180^\circ$
 (sum of two opposite angle in cyclic quadrilateral)
 $\Rightarrow \angle D = 180^\circ - \angle B$
 $= 180^\circ - 92^\circ = 88^\circ$
 $\Rightarrow \angle DAE = \angle D = 88^\circ$ [alternate angle]
 $\Rightarrow \angle FAD = 88^\circ + 20^\circ = 108^\circ$
 $\Rightarrow \angle BCD = \angle FAD = 108^\circ$
 [angles in alternate segments]
 $\Rightarrow \angle BCD = 108^\circ$
24. (d) As, $\angle B + \angle D = 180^\circ$
 and $\angle A + \angle C = 180^\circ$
 [sum of two opposite angle in a cyclic quadrilateral]
 $x + 10 + 5y + 5 = 180^\circ$
 $\Rightarrow x + 5y = 165^\circ \dots(i)$
 $\therefore \angle A + \angle C = 180^\circ$
 $\Rightarrow 2x + 4 + 4y - 4 = 180^\circ$
 $\Rightarrow 2x + 4y = 180^\circ \dots(ii)$

On solving, Eqs. (i) and (ii), we get x and y are 40° and 25° , respectively.

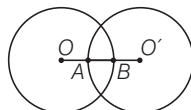
$$\therefore x + y = 40^\circ + 25^\circ = 65^\circ$$

25. (c) The locus of a point is the circumference of the circle with AB as diameter.



26. (b) As, the tangents drawn to a circle from a point outside it are equal. We have,
 $AD = AE$, $BD = BF$ and $CE = CF$
 $\Rightarrow AD = AB + BD = AB + BF$
 and $AD = AC + EC = AC + CF$
 $\therefore 2AD = AB + AC + (BF + CF)$
 $2AD = AB + AC + BC$

27. (d) $OB = 4$ cm



$$O'A = 2 \text{ cm}, OO' = 3 \text{ cm}$$

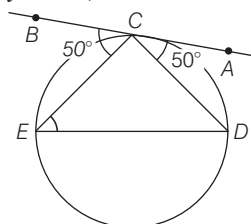
$$\text{As, } OO' \neq OB + O'A$$

So, circle does not touch each other externally.

$$\text{Also, } OO' \neq OB - O'A$$

So, circle does not touch internally, hence they intersect each other at two distinct points.

28. (a) Join ED , then



$$\angle DEC = \angle ACD = 50^\circ$$

[angle in alternate segment]

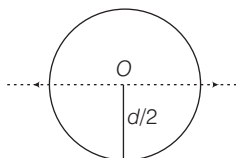
$$\angle EDC = \angle BCE = 50^\circ$$

[angle in alternate segment]

$$\therefore \angle DEC = \angle EDC$$

$$\text{So, } CD = CE$$

29. (b)



The locus of the centre of the front wheel of the cycle is a line parallel to the path of the wheel of the bicycle at a height $d/2$ cm.

30. (c) $\therefore \angle OAC = \angle OCA$
 [angle opposite to equal sides]
 $\therefore 2\angle OAC = 80^\circ$
 [external angle of $\triangle OAC$]
 $\angle OAC = \frac{80^\circ}{2} = 40^\circ \Rightarrow x = 40^\circ$

Hence, the value of x is 40° .

31. (a) Since, the direct common tangents of two circles divides the line joining their centres externally in the ratio of their radii. Here, both the circles being of equal radii. Hence, their ratio is $1 : 1$.

32. (a) Let $AB = 9$ cm, $BC = 7$ cm,

$$AC = 6$$
 cm

Let x, y and z be radii of circles with centre A, B, C .

$$x + y = 9 \dots(i)$$

$$y + z = 7 \dots(ii)$$

$$\text{and } z + x = 6 \dots(iii)$$

On adding Eqs. (i), (ii) and (iii), we get

$$2(x + y + z) = 22$$

$$\Rightarrow (x + y + z) = 11 \dots(iv)$$

On putting $x + y = 9$ in Eq. (iv), we get

$$9 + z = 11 \Rightarrow z = 2 \text{ cm}$$

On putting value of z in Eq. (ii), we get

$$y + 2 = 7 \Rightarrow y = 5$$

Again, putting value of y in Eq. (i), we get

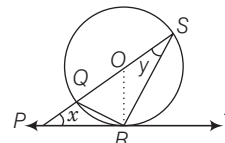
$$x + 5 = 9 \Rightarrow x = 4 \text{ cm}$$

So, radii are 4 cm, 5 cm and 2 cm.

33. (b) $\angle QRS = 90^\circ$ [angle in semi-circle]
 and $\angle QRP = \angle QSR = y^\circ$

[angle in alternate segments]

$$\text{Also, } \angle PRS = 90 + y^\circ$$



In $\triangle PRS$,

$$\angle SRP + \angle RPS + \angle PSR = 180^\circ$$

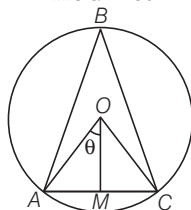
[$\therefore$ sum of all angles of triangle is 180°]

$$\Rightarrow (90 + y^\circ) + x^\circ + y^\circ = 180^\circ$$

$$\Rightarrow x + 2y^\circ = 90^\circ$$

34. (a) A line perpendicular to the given line, passing through the given point is the required locus.

35. (c) Given, $OA = 3$ cm and $AC = 3$ cm and $OA = OC = 3$ cm (radius of a circle)
In $\triangle AOC$, all sides are equal.
 $\therefore \triangle AOC$, is an equilateral triangle
 $\Rightarrow \angle AOC = 60^\circ$



Hence, $\angle ABC = \frac{1}{2} \angle AOC$

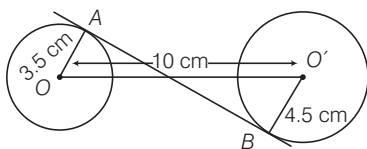
[Angle subtended by an arc at the centre is double the angle subtended by it at any point on the circle]
 $= \frac{60^\circ}{2} = 30^\circ$

36. (c) From figure, length of the transverse common tangent of these circles

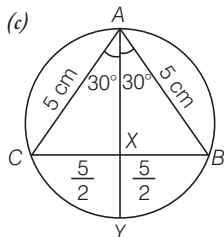
$$= \sqrt{(\text{distance between the centres of circles})^2 - (\text{sum of radius})^2}$$

$$= \sqrt{10^2 - (3.5 + 4.5)^2} = \sqrt{10^2 - 8^2}$$

$$= \sqrt{(100 - 64)} = \sqrt{36} = 6 \text{ cm}$$



37. (c)



In equilateral triangle, angle bisector and median are same

$$\therefore CX = BX = \frac{BC}{2} = \frac{5}{2} \text{ cm}$$

$$\text{In } \triangle CXA, \cos 30^\circ = \frac{AX}{AC} \Rightarrow \frac{\sqrt{3}}{2} = \frac{AX}{5}$$

$$\Rightarrow AX = \frac{\sqrt{3}}{2} \times 5 = \frac{5\sqrt{3}}{2} \text{ cm}$$

AY and BC are the chord of circle.

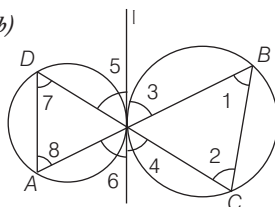
$$\therefore AX \cdot XY = BX \cdot XC$$

$$\Rightarrow \frac{5\sqrt{3}}{2} \cdot XY = \frac{5}{2} \cdot \frac{5}{2} \Rightarrow XY = \frac{5}{2\sqrt{3}}$$

$$\therefore AY \cdot AX = \left(\frac{5\sqrt{3}}{2} + \frac{5}{2\sqrt{3}} \right) \times \frac{5\sqrt{3}}{2}$$

$$= 25 \text{ cm}^2$$

38. (b)



l is the common tangent of two circles

Then, $\angle 1 = \angle 4$ and $\angle 5 = \angle 8$

[angle in alternate segment]

But $\angle 4 = \angle 5$ [opposite angles]

$$\therefore \angle 1 = \angle 8$$

Similarly $\angle 2 = \angle 7$

But these are alternate angles

$\Rightarrow AD \parallel BC \Rightarrow ACBD$ is a trapezium.

39. (c) We know that, two tangents drawn from an external point to a circle are equal in length.

$$\therefore AP = AS \quad \dots(i)$$

$$BP = BQ \quad \dots(ii)$$

$$CR = CQ \quad \dots(iii)$$

$$DR = DS \quad \dots(iv)$$

On adding Eqs. (i), (ii), (iii) and (iv), we get

$$(AP + BP) + (CR + DR)$$

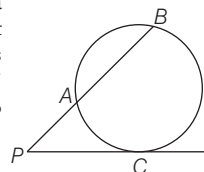
$$= (AS + DS) + (BQ + CQ)$$

$$\Rightarrow AB + CD = AD + BC$$

40. (a) If a secant to a circle intersect circle at points A and B and PC is a tangent to circle, then

$$PC^2 = PA \times PB$$

which is equivalent to saying that area of rectangle with PA and PB as sides is equal to the area of square with PC as sides.



41. (b) Given, $\angle BAD = 60^\circ$

and $\angle ADC = 105^\circ$

As ABCD is a cyclic quadrilateral

$$\therefore \angle DCP = \angle BAD = 60^\circ$$

and $\angle ADC + \angle CDP$

$$= 180^\circ \text{ [linear line]}$$

$$\Rightarrow 105^\circ + \angle CDP = 180^\circ$$

$$\Rightarrow \angle CDP = 75^\circ$$

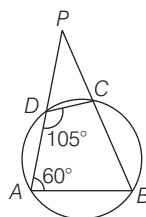
In $\triangle CPD$,

$$\angle DCP + \angle CDP + \angle DPC = 180^\circ$$

[by angle sum property of a triangle]

$$\Rightarrow 60^\circ + 75^\circ + \angle DPC = 180^\circ$$

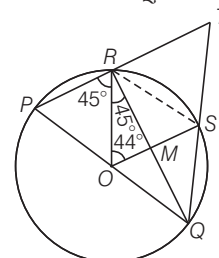
$$\Rightarrow \angle DPC = 180^\circ - 135^\circ = 45^\circ$$



42. (d) $\angle PRQ = 90^\circ$ as angle in a semi-circle is 90° .

Since, OR is a bisector of $\angle PRQ$.

$$\therefore \angle PRO = \angle ORQ = 45^\circ$$



Also, $OP = OR$ [radius]

$$\therefore \angle OPR = 45^\circ$$

in $\triangle ORS$, $OR = OS$

$$\Rightarrow \angle ORS = \angle OSR$$

in $\triangle ORS$, by angle sum property of a triangle,

$$\angle ORS + \angle OSR + \angle ROS = 180^\circ$$

$$2\angle ORS = 180^\circ - 44^\circ$$

$$2\angle ORS = 136^\circ$$

$$\therefore \angle ORS = \angle OSR = \frac{136^\circ}{2} = 68^\circ$$

$$\therefore \angle MRS = 68^\circ - 45^\circ = 23^\circ$$

$$\Rightarrow \angle PRS = 90^\circ + 23^\circ = 113^\circ$$

By properties of cyclic quadrilateral,

$$\angle PRS + \angle PQS = 180^\circ$$

$$\Rightarrow \angle PQS = 180^\circ - 113^\circ = 67^\circ$$

in $\triangle PTQ$,

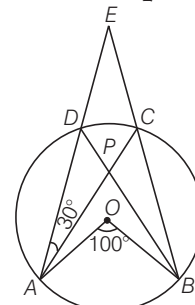
$$\angle QPT + \angle PQT + \angle PTQ = 180^\circ$$

[by angle sum property of a triangle]

$$\Rightarrow \angle PTQ = 180^\circ - 45^\circ - 67^\circ = 68^\circ$$

$$\therefore \angle RTS = \angle PTQ = 68^\circ$$

43. (b) Since, $\angle ADB = \frac{1}{2} \angle AOB = 50^\circ$



In $\triangle DPA$, by angle sum property of a triangle

$$\angle DAP + \angle ADP + \angle DPA = 180^\circ$$

$$\Rightarrow 30^\circ + 50^\circ + \angle DPA = 180^\circ$$

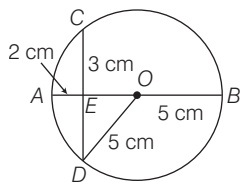
$$\Rightarrow \angle DPA = 100^\circ$$

Also, DPB is a straight line,

$$\therefore \angle DPA + \angle APB = 180^\circ$$

$$\Rightarrow \angle APB = 180^\circ - 100^\circ = 80^\circ$$

44. (b) In $\triangle OED$,



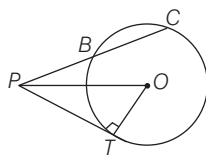
[by pythagoras theorem]

$$\begin{aligned}(OD)^2 &= (DE)^2 + (EO)^2 \\ \Rightarrow (5)^2 &= (DE)^2 + (3)^2 \\ \Rightarrow (DE)^2 &= 25 - 9 = 16 \\ \Rightarrow DE &= 4 \text{ cm}\end{aligned}$$

45. (a) Given, $PO = 10$ cm, $OT = 6$ cm and $PB = 5$ cm

In right angled $\triangle OTP$,

$$(OP)^2 = (PT)^2 + (OT)^2$$

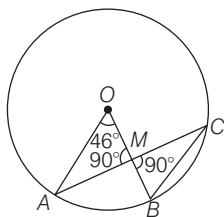


$$\begin{aligned}\Rightarrow (10)^2 &= (PT)^2 + 6^2 \\ \Rightarrow PT^2 &= 100 - 36 \Rightarrow PT^2 = 64 \\ \Rightarrow PT &= \sqrt{64} \\ \therefore PT &= 8 \text{ cm}\end{aligned}$$

By using theorem of circle,

$$\begin{aligned}(PT)^2 &= PB \times PC \\ \Rightarrow 8^2 &= 5 \times (BC + PB) \\ \Rightarrow 64 &= 5(BC + 5) \Rightarrow 5BC = 39 \\ \therefore BC &= 7.8 \text{ cm}\end{aligned}$$

46. (c) Given, $\angle AOB = 46^\circ$,



$$\therefore \angle ACB = \frac{1}{2} \angle AOB = \frac{1}{2} \times 46^\circ = 23^\circ$$

[angle subtended on the circumference is half of the angle subtend on centre.]

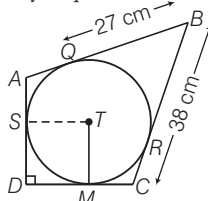
In $\triangle MCB$,

$$\angle C + \angle B + \angle BMC = 180^\circ$$

[by angle sum property of a triangle]

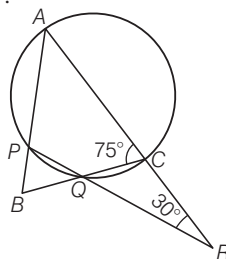
$$\begin{aligned}\Rightarrow 23^\circ + \angle B + 90^\circ &= 180^\circ \\ \Rightarrow \angle B &= 67^\circ\end{aligned}$$

47. (b) We know that, the tangents drawn to the circle from a point outside the circle are always equal.



$$\begin{aligned}\therefore BQ &= BR = 27 \text{ cm} \\ \Rightarrow RC &= BC - BR \\ &= 38 - 27 = 11 \text{ cm} \\ \therefore RC &= CM = 11 \text{ cm} \\ \text{Now, } DM &= 25 - 11 = 14 \text{ cm} \\ \therefore ST &= DM = 14 \text{ cm}\end{aligned}$$

48. (d) We know that, the sum of opposite angles in a cyclic quadrilateral is always 180° .



$$\begin{aligned}\angle ACQ + \angle APQ &= 180^\circ \\ \Rightarrow 75^\circ + \angle APQ &= 180^\circ \\ \Rightarrow \angle APQ &= 180^\circ - 75^\circ = 105^\circ \\ \text{Since, } \angle APQ + \angle BPQ &= 180^\circ\end{aligned}$$

[linear pair]

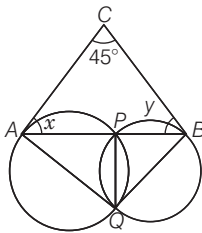
$$\begin{aligned}\therefore 105^\circ + \angle BPQ &= 180^\circ \\ \Rightarrow \angle BPQ &= 180^\circ - 105^\circ = 75^\circ \\ \text{Since, } \angle ACQ \text{ is an exterior angle of } \triangle RCQ.\end{aligned}$$

$$\begin{aligned}\therefore \angle ACQ &= \angle CRQ + \angle CQR \\ \Rightarrow 75^\circ &= 30^\circ + \angle CQR \\ \Rightarrow \angle CQR &= 45^\circ = \angle PQB\end{aligned}$$

[vertically opposite angles]

$$\begin{aligned}\text{In } \triangle BPQ, \angle B + \angle P + \angle Q &= 180^\circ \\ \text{[by angle sum property of a triangle]} \\ \Rightarrow \angle B + 75^\circ + 45^\circ &= 180^\circ \Rightarrow \angle B = 60^\circ\end{aligned}$$

49. (d)



$$\begin{aligned}\text{In } \triangle CAB, 45^\circ + \angle x + \angle y &= 180^\circ \\ \text{[by angle sum property of triangle]}\end{aligned}$$

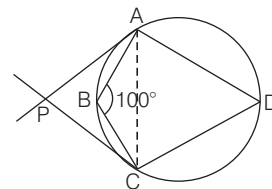
$$\Rightarrow \angle x + \angle y = 180^\circ - 45^\circ = 135^\circ \dots (i)$$

$$\angle x = \angle AQP \text{ and } \angle y = \angle BQP$$

[angles in alternate segment]

$$\begin{aligned}\therefore \angle AQB &= \angle AQP + \angle BQP \\ &= \angle x + \angle y = 135^\circ\end{aligned}$$

50. (b) We know that, the sum of opposite angles of a cyclic quadrilateral is always 180° .



$$\therefore \angle B + \angle D = 180^\circ$$

$$\Rightarrow 100 + \angle D = 180^\circ \Rightarrow \angle D = 80^\circ$$

$$\therefore \angle ACP = \angle PAC = 80^\circ$$

[by theorem of alternate interior segment]

In $\triangle PAC$,

$$\angle P + \angle PAC + \angle PCA = 180^\circ$$

[by angle sum property of a triangle]

$$\Rightarrow \angle P + 80^\circ + 80^\circ = 180^\circ$$

$$\Rightarrow \angle P = 180^\circ - 160^\circ = 20^\circ$$

51. (b) Since, the length of tangents from an external point to a circle are equal

$$\therefore CR = CS = 27 \text{ cm}$$

$$\therefore BR = (BC - CR) = 38 - 27 = 11 \text{ cm}$$

$$\Rightarrow BQ = BR = 11 \text{ cm}$$

join OQ, then PAQO is a rectangle.

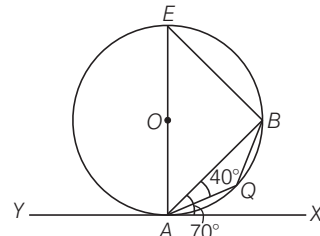
$$\therefore AQ = PQ = 10 \text{ cm}$$

$$\text{So, } AB = (AQ + BQ) = (10 + 11) = 21 \text{ cm}$$

52. (b) Given, $\angle BAX = 70^\circ$

$$\text{and } \therefore \angle BEA = 70^\circ$$

[angles in alternate segment]



Since, AQBE is a cyclic quadrilateral.

$$\therefore \angle AEB + \angle AQB = 180^\circ$$

$$\Rightarrow \angle AQB = 180^\circ - 70^\circ = 110^\circ$$

In $\triangle ABQ$,

$$\angle ABQ + \angle BAQ + \angle AQB = 180^\circ$$

$$\Rightarrow \angle ABQ = 180^\circ - (40^\circ + 110^\circ)$$

$$= 180^\circ - 150^\circ = 30^\circ$$